

Triangle Congruence Postulates and CPCTC

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Congruent triangles are triangles that have exactly the same size and shape.

SSS Congruence Postulate

The **SSS Congruence Postulate** states that if the three sides of one triangle are congruent to the three corresponding sides of another triangle, then the triangles are congruent.

SSS

Three corresponding sides are congruent.

SAS Congruence Postulate

The **SAS Congruence Postulate** states that if two sides and the included angle of one triangle are congruent to the corresponding parts of another triangle, then the triangles are congruent.

SAS

Two sides and the included angle are congruent.

ASA Congruence Postulate

The **ASA Congruence Postulate** states that if two angles and the included side of one triangle are congruent to the corresponding parts of another triangle, then the triangles are congruent.

ASA

Two angles and the included side are congruent.

AAS Congruence Theorem

The **AAS Congruence Theorem** states that if two angles and a non-included side of one triangle are congruent to the corresponding parts of another triangle, then the triangles are congruent.

AAS

Two angles and a non-included side are congruent.

HL Congruence Theorem

The **HL Congruence Theorem** applies only to right triangles.

If the hypotenuse and one leg of a right triangle are congruent to those of another right triangle, then the triangles are congruent.

HL

Hypotenuse and one leg are congruent.

CPCTC

CPCTC means Corresponding Parts of Congruent Triangles are Congruent.

After proving two triangles congruent, all corresponding sides and angles are also congruent.

Examples

Example 1

In $\triangle ABC$ and $\triangle DEF$,

$$AB = DE, \quad BC = EF, \quad AC = DF$$

Which congruence postulate proves the triangles congruent?

SSS

Example 2

In $\triangle PQR$ and $\triangle XYZ$,

$$PQ = XY, \quad \angle P = \angle X, \quad PR = XZ$$

Identify the congruence postulate.

SAS

Example 3

In $\triangle MNO$ and $\triangle RST$,

$$\angle M = \angle R, \quad MN = RS, \quad \angle N = \angle S$$

What congruence postulate applies?

ASA

Example 4

In $\triangle GHI$ and $\triangle JKL$,

$$\angle G = \angle J, \quad \angle I = \angle L, \quad HI = KL$$

Which theorem proves the triangles congruent?

AAS

Example 5

Two right triangles have hypotenuses measuring 13 cm and one corresponding leg measuring 5 cm.

Which theorem proves they are congruent?

HL

Example 6

Given

$$\triangle ABC \cong \triangle DEF$$

identify the congruent angle corresponding to $\angle B$.

$$\boxed{\angle E}$$

Example 7

Given

$$\triangle PQR \cong \triangle XYZ$$

identify the side corresponding to QR .

$$\boxed{YZ}$$

Practice Exercises

Practice 1

In $\triangle ABC$ and $\triangle DEF$,

$$AB = DE, \quad BC = EF, \quad AC = DF$$

Which congruence postulate proves the triangles congruent?

Practice 2

In $\triangle PQR$ and $\triangle XYZ$,

$$PQ = XY, \quad \angle P = \angle X, \quad PR = XZ$$

Identify the congruence postulate.

Practice 3

In $\triangle MNO$ and $\triangle RST$,

$$\angle M = \angle R, \quad MN = RS, \quad \angle N = \angle S$$

What congruence postulate applies?

Practice 4

In $\triangle GHI$ and $\triangle JKL$,

$$\angle G = \angle J, \quad \angle I = \angle L, \quad HI = KL$$

Which theorem proves the triangles congruent?

Practice 5

Two right triangles have hypotenuses measuring 13 cm and one corresponding leg measuring 5 cm.

Which theorem proves they are congruent?

Practice 6

Given

$$\triangle ABC \cong \triangle DEF$$

identify the congruent angle corresponding to $\angle B$.

Practice 7

Given

$$\triangle PQR \cong \triangle XYZ$$

identify the side corresponding to QR .

Activities

Activity 1

In $\triangle UVW$ and $\triangle XYZ$,

$$UV = XY, \quad VW = YZ, \quad UW = XZ$$

Determine the congruence postulate.

Activity 2

In $\triangle ABC$ and $\triangle DEF$,

$$AB = DE, \quad \angle B = \angle E, \quad BC = EF$$

Identify the congruence postulate.

Activity 3

In $\triangle JKL$ and $\triangle MNP$,

$$\angle J = \angle M, \quad JK = MN, \quad \angle K = \angle N$$

Determine the congruence postulate.

Activity 4

In $\triangle QRS$ and $\triangle TUV$,

$$\angle Q = \angle T, \quad \angle R = \angle U, \quad QS = TV$$

Identify the congruence theorem.

Activity 5

Two right triangles have hypotenuses measuring 17 cm and one corresponding leg measuring 8 cm.

Which theorem proves they are congruent?

Activity 6

Given

$$\triangle GHI \cong \triangle JKL$$

identify the congruent angle corresponding to $\angle H$.

Activity 7

Given

$$\triangle ABC \cong \triangle PQR$$

identify the side corresponding to AC .

Thank you!